TEXORA · LECTURE SHEET

The Auto Cone Winding Machine

The automatic cone winder (Autoconer): it takes the small, fault-carrying cops from the ring frame and builds large, clean, spliced cones for weaving, knitting and dyeing — clearing out the yarn's defects as it winds, at over a thousand metres a minute.

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1. Objectives of winding

- **Package building:** to transfer yarn from the small ring-frame cop (holding only about 60–100 g) into a large cone of 1.5–2.5 kg that can feed a warping machine, loom, knitting machine or dyeing package line without repeated piecing-up.
- **Yarn clearing:** to detect and remove the yarn's faults — thick places, thin places, neps, slubs and foreign matter — cutting them out as the yarn runs and re-joining the ends, so the delivered yarn meets a defined quality standard.
- **Joining without weakness:** to replace the knots of old winding with **splices** that are nearly invisible and nearly as strong as the yarn itself.
- **Controlled package formation:** to wind at controlled tension and density, with a package shape (cone taper, traverse pattern) that will unwind smoothly at high speed, and — for dyeing packages — at the low, uniform density dye liquor requires.

2. Working principle

Each winding head works independently and automatically through the same cycle. Yarn unwinds from a cop in the head's magazine-fed creel, rises through a **tensioning device** that holds a steady, adjustable winding tension, and passes an **electronic yarn clearer** that measures the running yarn continuously. It then travels through the **splicing zone** (kept ready, not normally in use) and is wound onto the cone. The cone is driven by friction from a rotating **drum**: a grooved drum both rotates the package and guides the yarn in its helical grooves, traversing it back and forth to build the cone; on precision-winding machines a separate traversing guide and a positively driven spindle replace the grooved drum. When the clearer detects a fault beyond its settings — or the cop runs out, or an end breaks — the head stops, cuts the yarn, sucks the two ends back, and the **splicer** joins them pneumatically; the head then restarts itself. A magazine of cops, automatic cop feed, and a doffer that removes full cones let one operator tend scores of heads.

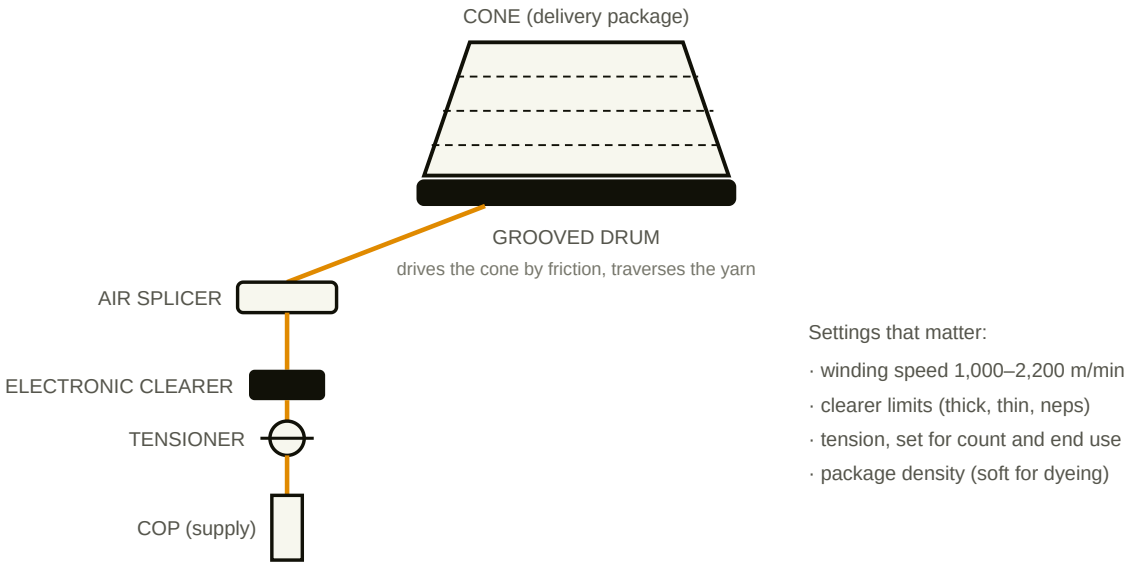


Figure 1 — Yarn path through one winding head: cop → tensioner → electronic clearer → air splicer (in reserve) → cone, rotated and traversed by the grooved drum. One head of many; each cycle is automatic.

3. Main parts and their functions

PART	FUNCTION
Cop magazine and feed	Holds several cops per head and presents a fresh cop automatically when one empties; the tail end is prepared for splicing into the running yarn
Tension device	Disc, gate or electronically controlled tensioner: sets the winding tension that decides package density and prevents both slack coils (too low) and stretched, broken ends (too high)
Electronic yarn clearer	Capacitive or optical sensor measuring yarn diameter/mass continuously; signals a cut when a fault exceeds the set limits for thick place, thin place, nep or foreign fibre; modern clearers also classify and count faults (e.g. the class-matrix of yarn clearers)
Cutter and suction	On a clearer signal, cuts the yarn and sucks both ends to the splicer; also removes the cut-out fault length so it can never reach the cone
Air splicer	Untwists the two end-fibres with air jets and re-twists them together: a joint of near-parent strength, smooth enough to pass heddles and needles. (See Section 4.)
Knotter (older/simpler heads)	Ties a mechanical knot instead of splicing — cheaper, but the knot is bulky, weaker in appearance and can jam or mark fabric
Waxing unit (for knitting yarns)	Rotating wax block the yarn rubs against, laying a film of paraffin that lowers friction in knitting

Grooved winding drum	Drives the cone by surface friction; its helical grooves traverse the yarn to build the package; groove pitch and drum design set the traverse pattern and help prevent ribboning (pattern winding)
Package cradle and doffer	Holds the cone under controlled pressure against the drum; automatic doffing ejects full cones onto a conveyor

4. Splicer versus knotter

A **knot** concentrates the joint into a hard lump roughly two to three yarn diameters wide with protruding tails: it can catch in guides and heddle eyes, shows in fine fabrics, and is a deliberate weak irregularity. An **air splice** interleaves and entangles the fibres of the two ends with opposed air blasts, so the joint has close to the parent yarn's diameter, no tails, and typically 75–95 % of its strength. That is why quality winding abandoned knots: for weaving and knitting yarns the splicer is standard, and knotters survive mainly where joints hardly matter (some coarse or technical yarns). Splice quality is itself a setting — splicing air pressure and duration are tuned per fibre and count, and splices are checked by deliberately breaking test joints.

5. Settings and winding quality

- **Winding speed:** 1,000–2,200 m/min is available; the used value balances production against clearer performance, splice rate and package quality — finer and weaker yarns run slower.
- **Clearer settings:** set from the fabric quality required, not from what the yarn happens to contain. Tight settings give clean yarn but many cuts — each cut is a splice, a stop, and a small efficiency loss; loose settings pass faults into the fabric.
- **Tension:** higher tension → denser, harder cone (good for storage and warping); dye packages are wound deliberately soft and at low tension so liquor penetrates evenly.
- **Anti-patterning:** when successive traverse cycles coincide, coils stack into ribbons that later collapse and break at unwinding; drum groove design and electronic anti-patterning vary the traverse to prevent it.

WORKED EXAMPLE — WINDING HEAD PRODUCTION

A head winds 30 Ne yarn at 1,200 m/min with a winding efficiency of 70 % (stoppage time for clearing cuts, splices and cop changes).

Length wound = 1,200 × 60 = 72,000 m/h ≈ 78,740 yd/h.

30 Ne contains 25,200 yd per pound, so theoretical output = 78,740 ÷ 25,200 ≈ 3.12 lb/h ≈ 1.42 kg/h.

> At 70 % efficiency: ≈ **0.99 kg/h per head** — a useful reminder that clearing quality is bought with winding efficiency.

Glossary

Autoconer — automatic cone winding machine. **Cone** — the conical delivery package built by the winder. **Yarn clearer** — the sensor-and-cutter system that removes out-of-limit faults. **Splice** — a pneumatic fibre-entangled joint replacing a knot. **Knotter** — the mechanical tying device splicers replaced. **Grooved drum** — the friction drive drum whose grooves traverse the yarn. **Ribboning (patterning)** — coils stacking in regular bands, causing collapse at unwinding. **Waxing** — paraffin application to knitting yarns to lower friction.

Key points

• The auto coner does three jobs at once: builds a big package, clears faults to a set standard, and joins ends by splicing.
• Yarn path: cop → tensioner → clearer → splicer zone → drum-driven cone.
• Clearer settings are a fabric-quality decision: every removed fault costs a stop, a splice and efficiency.
• Air splices reach roughly 75–95 % of yarn strength with no bulky knot; knotters remain only for coarse, undemanding work.
• Package density is set by tension: hard for warping, soft for dyeing.
• Production per head = speed × count conversion × efficiency — worked example gives ≈ 1 kg/h for 30 Ne at 1,200 m/min.

Practice

1. Calculate the production (kg/h) of one head winding 20 Ne at 1,500 m/min and 75 % efficiency.
2. A weaving mill complains of knot marks in fine shirting although the winder "joins every break". Diagnose the cause and the fix.
3. Why must the fault length cut out by the clearer be positively removed by suction rather than allowed to fall or be pushed aside?
4. Explain the trade-off a mill manager faces when tightening clearer settings, in terms of fabric quality, splicer load and winding efficiency.
5. Why are dye packages wound at lower tension than warping packages, and what defect appears in dyeing if they are not?

Answer key

1. Length = 1,500 × 60 = 90,000 m/h ≈ 98,425 yd/h. 20 Ne = 16,800 yd/lb, so output = 98,425 ÷ 16,800 ≈ 5.86 lb/h ≈ 2.66 kg/h; × 0.75 ≈ **1.99 kg/h per head**.
2. The heads are using knotters: each knot is a hard lump that shows through fine fabric and can catch in the heddles. The fix is to fit/use air splicers (correct pressure and duration for the count) so joints are smooth and near parent diameter.
3. The cut fault is exactly the material that must never be sold: if it merely fell aside it could be dragged back onto the package or a neighbouring head's cone by air currents or handling. Suction guarantees the rejected length leaves the process entirely, and it also delivers both yarn ends to the splicer in a controlled way.
4. Tighter settings remove more faults, so fabric quality (fewer bars, holes and visible defects) rises. But each removal stops the head and consumes a splice cycle: splices per cone rise, efficiency falls, and the splices themselves become numerous enough to matter. The optimum is set by the fabric's quality class, not by the clearer's maximum sensitivity.
5. Dye liquor must flow uniformly through the package; a hard, dense package resists flow at its core, producing unlevel dyeing (dark outside, pale inside). Low, even winding tension keeps the package permeable so the whole mass dyes to one shade.

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